

---

# Neutrinos via Charm Decays in Astrophysical Sources

TALK IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE DEGREE OF BACHELOR OF SCIENCE

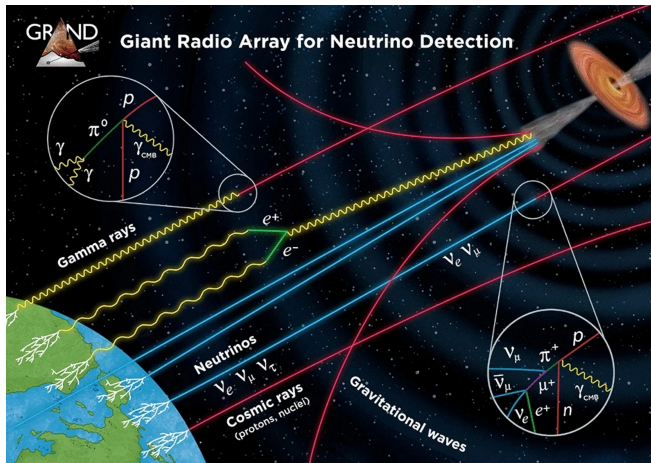
---

Fritz Ali Agildere

**September 3, 2024**

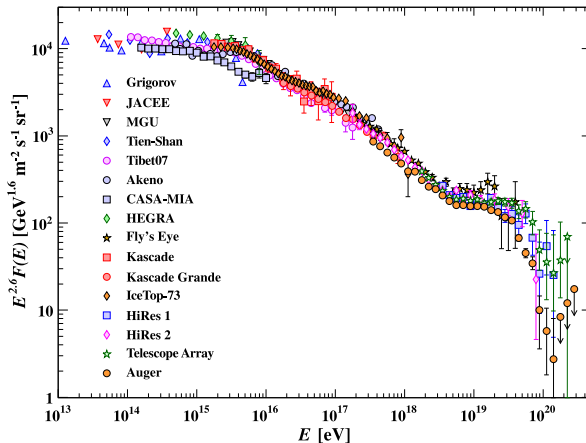
Technical University of Dortmund & Ruhr University Bochum

## Multimessenger Astrophysics



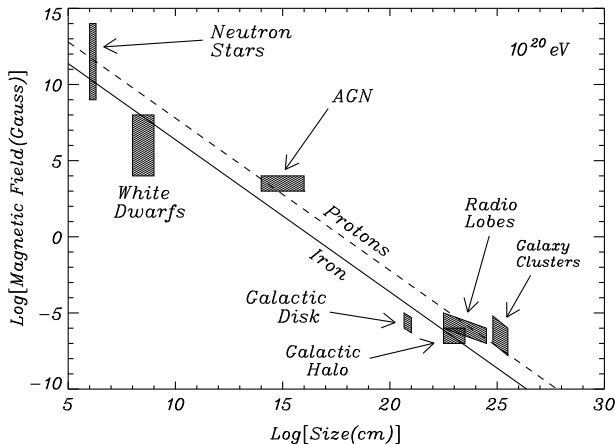
X. L. Qian et al.  
*Nucl. Sci. Tech.* (2021)

## Cosmic Ray Spectrum & Cutoff



K. A. Olive et al.  
*Chin. Phys. C* (2014)

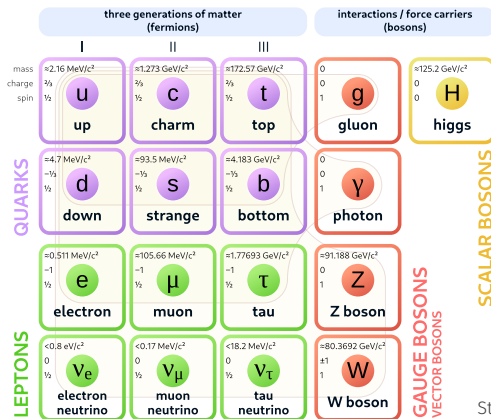
## Plausible Cosmic Ray Origins



A. V. Olinto  
AIP Conf. Proc. (2000)

## Elementary Particles & Hadrons

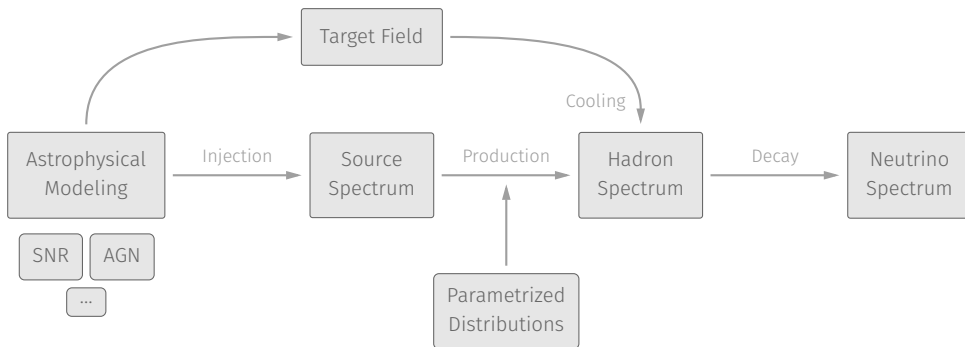
### Standard Model of Elementary Particles

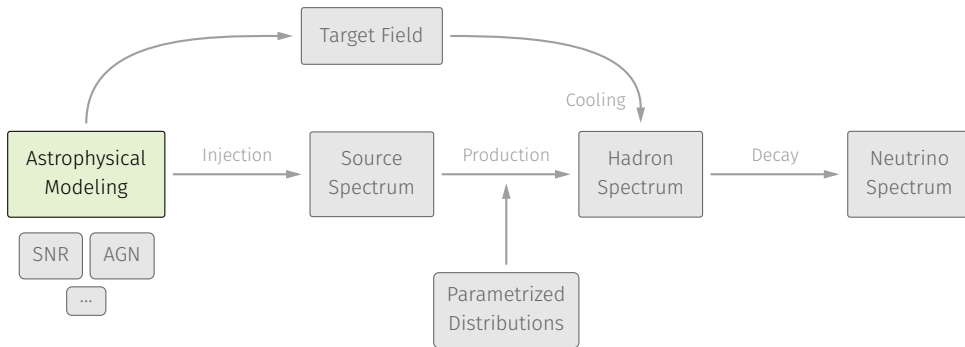


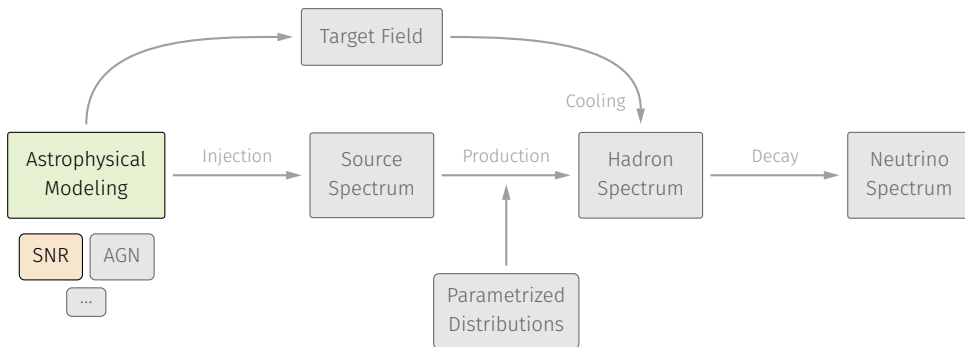
Standard Model  
Wikimedia Commons (2019)

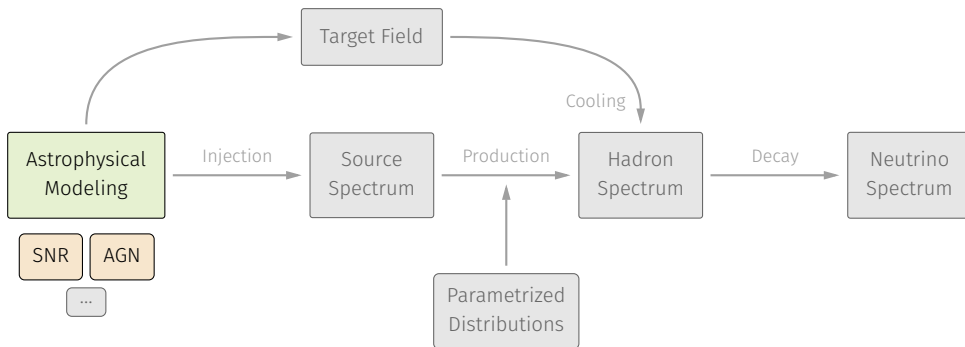
## Relevant Hadron Properties

| $h$           | $q_h$      | $\tau_h / \text{s}$   | $m_h / \text{GeV}$ | $s_h / \text{GeV}^2$ | $B_{h \rightarrow \nu}$ |
|---------------|------------|-----------------------|--------------------|----------------------|-------------------------|
| $p$           | $uud$      | $\infty$              | 0.94               | 15.98                | 0.000                   |
| $\pi^+$       | $u\bar{d}$ | $26.0 \times 10^{-9}$ | 0.14               | 10.23                | 1.000                   |
| $K^+$         | $u\bar{s}$ | $12.4 \times 10^{-9}$ | 0.49               | 12.62                | 0.636                   |
| $D_0$         | $c\bar{u}$ | $0.4 \times 10^{-12}$ | 1.86               | 0.45                 | 0.067                   |
| $D^+$         | $c\bar{d}$ | $1.0 \times 10^{-12}$ | 1.87               | 0.40                 | 0.176                   |
| $D_s^+$       | $c\bar{s}$ | $0.5 \times 10^{-12}$ | 1.97               | 0.71                 | 0.065                   |
| $\Lambda_c^+$ | $udc$      | $0.2 \times 10^{-12}$ | 2.29               | 1.61                 | 0.045                   |

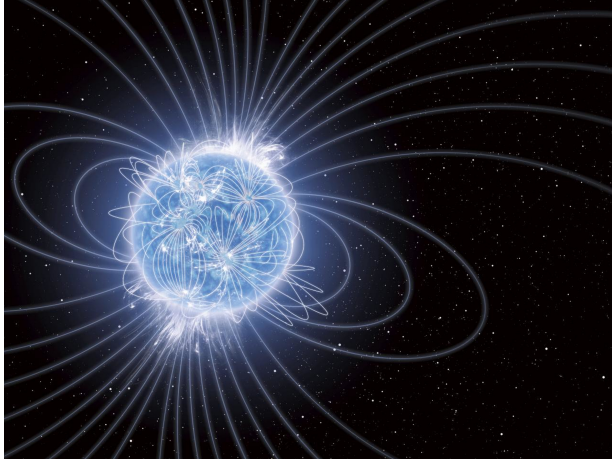






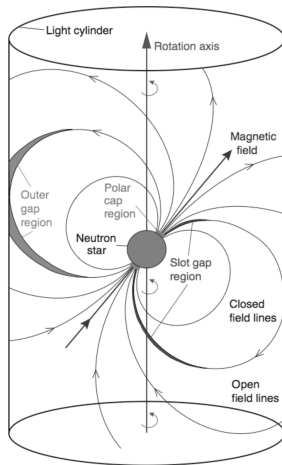


## Young Neutron Stars



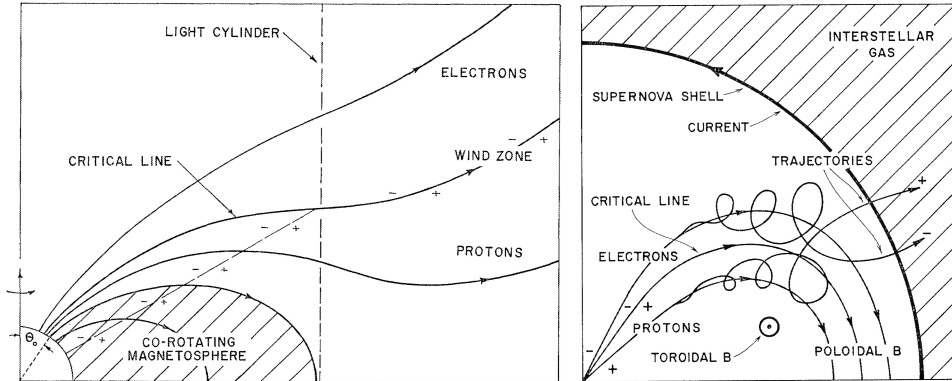
R. Takeshige  
*Magnetar* (2020)

## Dipolar Pulsar Model



E. Aliu et al.  
*Science* (2008)

## Particle Acceleration



P. Goldreich, W. H. Julian *ApJ* (1969)



## Magnetar Spindown

- Electromagnetic Luminosity

## Magnetar Spindown

### ■ Electromagnetic Luminosity

$$L = \frac{B_{ns}^2 R_{ns}^6 \omega_0^4 (1 + \sin^2 \chi)}{4c^3 (1 + t/t_{sd})^2}$$

## Magnetar Spindown

### ■ Electromagnetic Luminosity

$$L = \frac{B_{ns}^2 R_{ns}^6 \omega_0^4 (1 + \sin^2 \chi)}{4c^3 (1 + t/t_{sd})^2}$$

## Magnetar Spindown

### ■ Electromagnetic Luminosity

$$L = \frac{B_{ns}^2 R_{ns}^6 \omega_0^4 (1 + \sin^2 \chi)}{4c^3 (1 + t/t_{sd})^2}$$

## Magnetar Spindown

### ■ Electromagnetic Luminosity

$$L = \frac{B_{ns}^2 R_{ns}^6 \omega_0^4 (1 + \sin^2 \chi)}{4c^3 (1 + t/t_{sd})^2}$$

## Magnetar Spindown

### ■ Electromagnetic Luminosity

$$L = \frac{B_{ns}^2 R_{ns}^6 \omega_0^4 (1 + \sin^2 \chi)}{4c^3 (1 + t/t_{sd})^2}$$

## Magnetar Spindown

- Electromagnetic Luminosity

$$L = \frac{B_{ns}^2 R_{ns}^6 \omega_0^4 (1 + \sin^2 \chi)}{4c^3 (1 + t/t_{sd})^2}$$

- Characteristic Timescale

## Magnetar Spindown

### ■ Electromagnetic Luminosity

$$L = \frac{B_{ns}^2 R_{ns}^6 \omega_0^4 (1 + \sin^2 \chi)}{4c^3 (1 + t/t_{sd})^2}$$

### ■ Characteristic Timescale

$$t_{sd} = \frac{2I_{ns} c^3}{B_{ns}^2 R_{ns}^6 \omega_0^2 (1 + \sin^2 \chi)}$$

## Magnetar Spindown

- Electromagnetic Luminosity

$$L = \frac{B_{ns}^2 R_{ns}^6 \omega_0^4 (1 + \sin^2 \chi)}{4c^3 (1 + t/t_{sd})^2}$$

- Characteristic Timescale

$$t_{sd} = \frac{2I_{ns} c^3}{B_{ns}^2 R_{ns}^6 \omega_0^2 (1 + \sin^2 \chi)}$$

- Monotonic Energy

## Magnetar Spindown

### ■ Electromagnetic Luminosity

$$L = \frac{B_{ns}^2 R_{ns}^6 \omega_0^4 (1 + \sin^2 \chi)}{4c^3 (1 + t/t_{sd})^2}$$

### ■ Characteristic Timescale

$$t_{sd} = \frac{2I_{ns} c^3}{B_{ns}^2 R_{ns}^6 \omega_0^2 (1 + \sin^2 \chi)}$$

### ■ Monotonic Energy

$$E^M = \frac{\eta e B_{ns} R_{ns}^3 \omega_0^2}{2c^2 (1 + t/t_{sd})}$$

## Magnetar Spindown

### ■ Electromagnetic Luminosity

$$L = \frac{B_{ns}^2 R_{ns}^6 \omega_0^4 (1 + \sin^2 \chi)}{4c^3 (1 + t/t_{sd})^2}$$

### ■ Characteristic Timescale

$$t_{sd} = \frac{2I_{ns} c^3}{B_{ns}^2 R_{ns}^6 \omega_0^2 (1 + \sin^2 \chi)}$$

### ■ Monotonic Energy

$$E^M = \frac{\eta e B_{ns} R_{ns}^3 \omega_0^2}{2c^2 (1 + t/t_{sd})}$$

## Magnetar Spindown

### ■ Electromagnetic Luminosity

$$L = \frac{B_{ns}^2 R_{ns}^6 \omega_0^4 (1 + \sin^2 \chi)}{4c^3 (1 + t/t_{sd})^2}$$

### ■ Characteristic Timescale

$$t_{sd} = \frac{2I_{ns} c^3}{B_{ns}^2 R_{ns}^6 \omega_0^2 (1 + \sin^2 \chi)}$$

### ■ Monotonic Energy

$$E^M = \frac{\eta e B_{ns} R_{ns}^3 \omega_0^2}{2c^2 (1 + t/t_{sd})}$$

## Magnetar Spindown

### ■ Electromagnetic Luminosity

$$L = \frac{B_{ns}^2 R_{ns}^6 \omega_0^4 (1 + \sin^2 \chi)}{4c^3 (1 + t/t_{sd})^2}$$

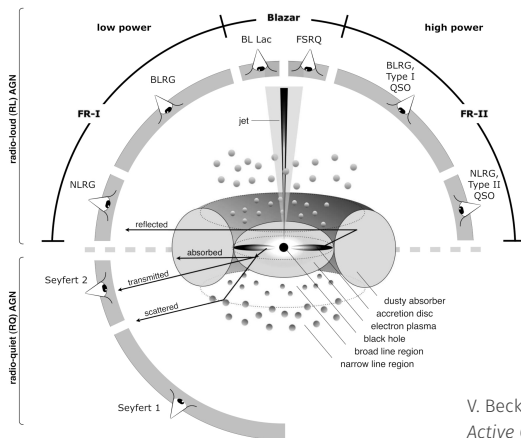
### ■ Characteristic Timescale

$$t_{sd} = \frac{2I_{ns} c^3}{B_{ns}^2 R_{ns}^6 \omega_0^2 (1 + \sin^2 \chi)}$$

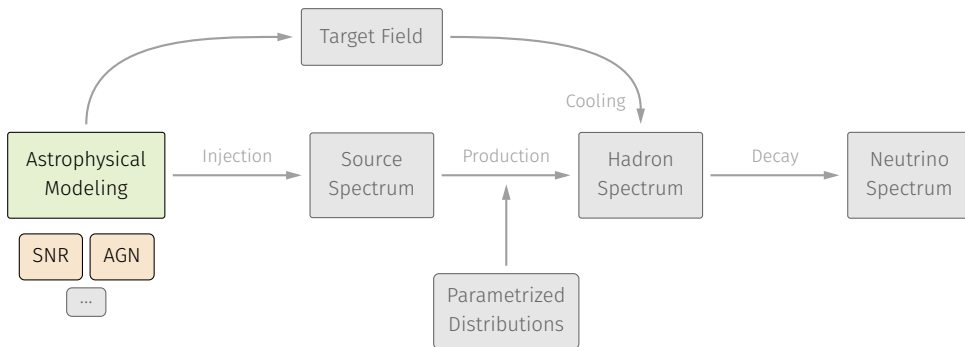
### ■ Monotonic Energy

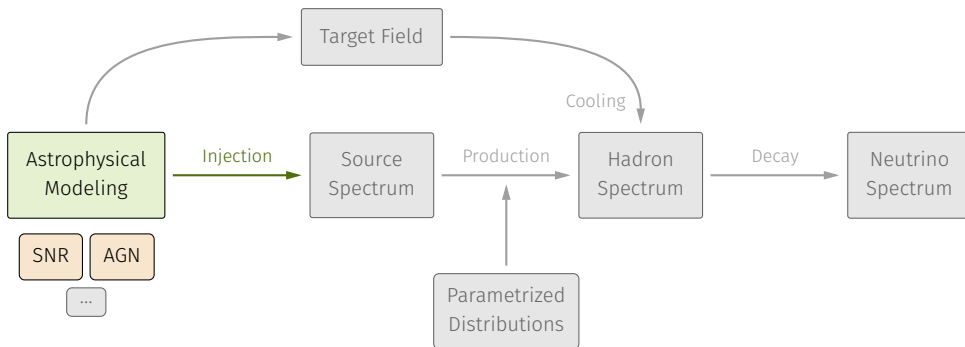
$$E^M = \frac{\eta e B_{ns} R_{ns}^3 \omega_0^2}{2c^2 (1 + t/t_{sd})}$$

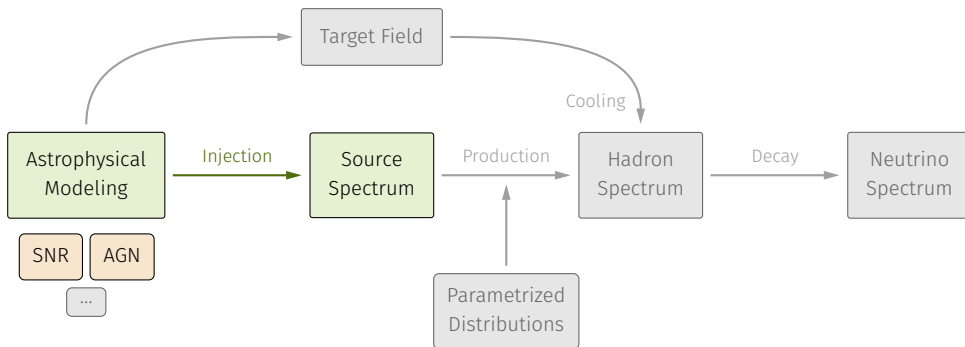
## Active Galactic Nuclei



V. Beckmann, C. Shrader  
*Active Galactic Nuclei* (2012)









## Injected Primary Spectrum

- Monochromatic Proton Energy

## Injected Primary Spectrum

### ■ Monochromatic Proton Energy

$$\frac{d\dot{N}_p}{dE_p} = \frac{B_{ns} R_{ns}^3 \omega_0^2}{ec(1 + t/t_{sd})} \delta(E_p - E^M)$$

## Injected Primary Spectrum

### ■ Monochromatic Proton Energy

$$\frac{d\dot{N}_p}{dE_p} = \frac{B_{ns} R_{ns}^3 \omega_0^2}{ec(1 + t/t_{sd})} \delta(E_p - E^M)$$

## Injected Primary Spectrum

### ■ Monochromatic Proton Energy

$$\frac{d\dot{N}_p}{dE_p} = \frac{B_{ns} R_{ns}^3 \omega_0^2}{ec(1 + t/t_{sd})} \delta(E_p - E^M)$$

## Injected Primary Spectrum

### ■ Monochromatic Proton Energy

$$\frac{d\dot{N}_p}{dE_p} = \frac{B_{ns} R_{ns}^3 \omega_0^2}{ec(1 + t/t_{sd})} \delta(E_p - E^M)$$

## Injected Primary Spectrum

### ■ Monochromatic Proton Energy

$$\frac{d\dot{N}_p}{dE_p} = \frac{B_{ns} R_{ns}^3 \omega_0^2}{ec(1 + t/t_{sd})} \delta(E_p - E^M)$$

$$\frac{dN_p}{dE_p} \propto E_p^{-1}$$

## Injected Primary Spectrum

### ■ Monochromatic Proton Energy

$$\frac{d\dot{N}_p}{dE_p} = \frac{B_{ns} R_{ns}^3 \omega_0^2}{ec(1 + t/t_{sd})} \delta(E_p - E^M)$$

$$\frac{dN_p}{dE_p} \propto E_p^{-1}$$

### ■ Stochastic Acceleration Power Law

## Injected Primary Spectrum

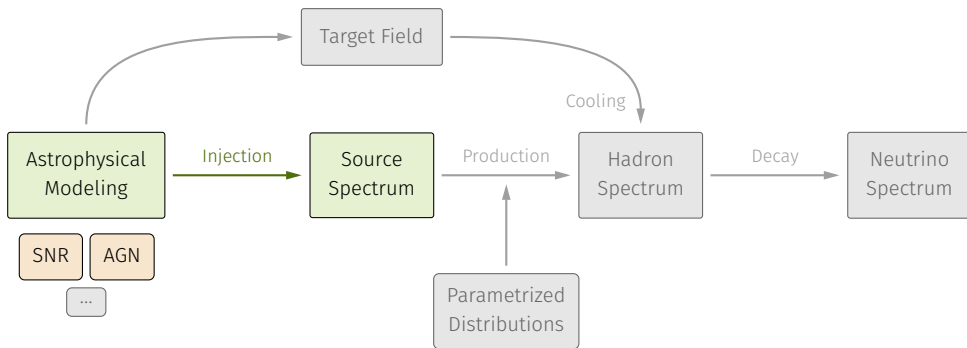
### ■ Monochromatic Proton Energy

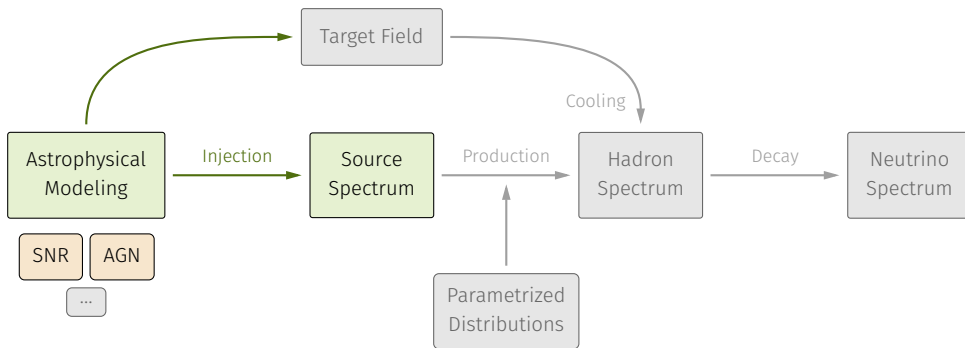
$$\frac{d\dot{N}_p}{dE_p} = \frac{B_{ns} R_{ns}^3 \omega_0^2}{ec(1 + t/t_{sd})} \delta(E_p - E^M)$$

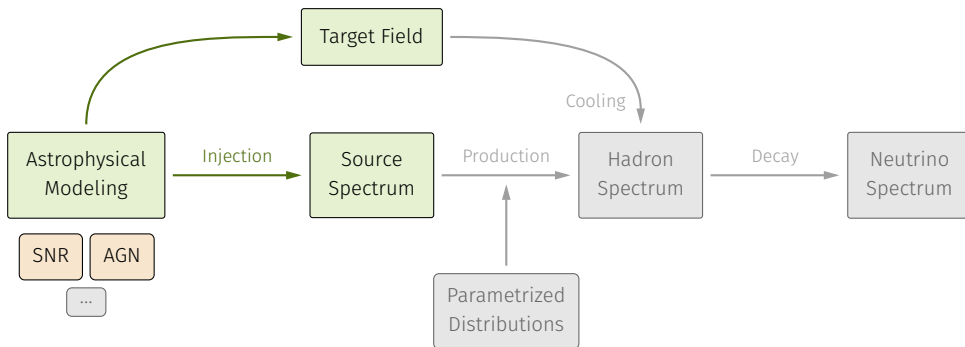
$$\frac{dN_p}{dE_p} \propto E_p^{-1}$$

### ■ Stochastic Acceleration Power Law

$$\frac{dN_p}{dE_p} \propto E_p^{-2}$$









## Magnetar Environment

- Supernova Shell Radius

## Magnetar Environment

- Supernova Shell Radius

$$R_{ej} = \beta_{ej} ct$$

## Magnetar Environment

- Supernova Shell Radius

$$R_{ej} = \beta_{ej} ct$$

- Ejecta Number Density

## Magnetar Environment

- Supernova Shell Radius

$$R_{ej} = \beta_{ej} ct$$

- Ejecta Number Density

$$n_{ej} = \frac{3M_{ej}}{4\pi R_{ej}^3 m_p}$$

## Magnetar Environment

- Supernova Shell Radius

$$R_{ej} = \beta_{ej} ct$$

- Ejecta Number Density

$$n_{ej} = \frac{3M_{ej}}{4\pi R_{ej}^3 m_p}$$

## Magnetar Environment

- Supernova Shell Radius

$$R_{ej} = \beta_{ej} ct$$

- Ejecta Number Density

$$n_{ej} = \frac{3M_{ej}}{4\pi R_{ej}^3 m_p}$$

## Magnetar Environment

- Supernova Shell Radius

$$R_{ej} = \beta_{ej} ct$$

- Ejecta Number Density

$$n_{ej} = \frac{3M_{ej}}{4\pi R_{ej}^3 m_p}$$

- Numerical Setup

## Magnetar Environment

### ■ Supernova Shell Radius

$$R_{ej} = \beta_{ej} ct$$

### ■ Ejecta Number Density

$$n_{ej} = \frac{3M_{ej}}{4\pi R_{ej}^3 m_p}$$

### ■ Numerical Setup

$$\begin{array}{llll} I_{ns} = 10^{45} \text{ g cm}^2 & R_{ns} = 10^6 \text{ cm} & B_{ns} = 10^{14.5} \text{ G} & t_{sd} = 10^{3.5} \text{ s} \\ M_{ej} = 10 M_{\odot} & \beta_{ej} = 0.1 & \eta = 0.1 & \end{array}$$



## Accretion Disk

- Thin Limit Scale Height

## Accretion Disk

### ■ Thin Limit Scale Height

$$h^2 = \frac{u_{sn}^2 R_{dc}^3}{GM_{bh}}$$

## Accretion Disk

- Thin Limit Scale Height

$$h^2 = \frac{u_{sn}^2 R_{dc}^3}{GM_{bh}}$$

- Gaussian Density Profile

## Accretion Disk

- Thin Limit Scale Height

$$h^2 = \frac{u_{sn}^2 R_{dc}^3}{GM_{bh}}$$

- Gaussian Density Profile

$$\rho \propto \exp\left(-\frac{GM_{bh} z^2}{2u_{sn}^2 R_{dc}^3}\right)$$

## Accretion Disk

- Thin Limit Scale Height

$$h^2 = \frac{u_{sn}^2 R_{dc}^3}{GM_{bh}}$$

- Gaussian Density Profile

$$\rho \propto \exp\left(-\frac{GM_{bh} z^2}{2u_{sn}^2 R_{dc}^3}\right)$$

## Accretion Disk

- Thin Limit Scale Height

$$h^2 = \frac{u_{sn}^2 R_{dc}^3}{GM_{bh}}$$

- Gaussian Density Profile

$$\rho \propto \exp\left(-\frac{GM_{bh} z^2}{2u_{sn}^2 R_{dc}^3}\right)$$

## Accretion Disk

- Thin Limit Scale Height

$$h^2 = \frac{u_{sn}^2 R_{dc}^3}{GM_{bh}}$$

- Gaussian Density Profile

$$\rho \propto \exp\left(-\frac{GM_{bh} z^2}{2u_{sn}^2 R_{dc}^3}\right)$$

- Parameter Choice

## Accretion Disk

### ■ Thin Limit Scale Height

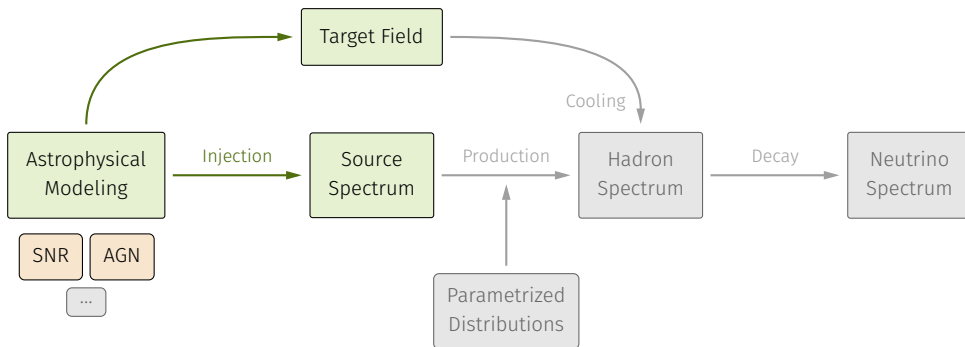
$$h^2 = \frac{u_{sn}^2 R_{dc}^3}{GM_{bh}}$$

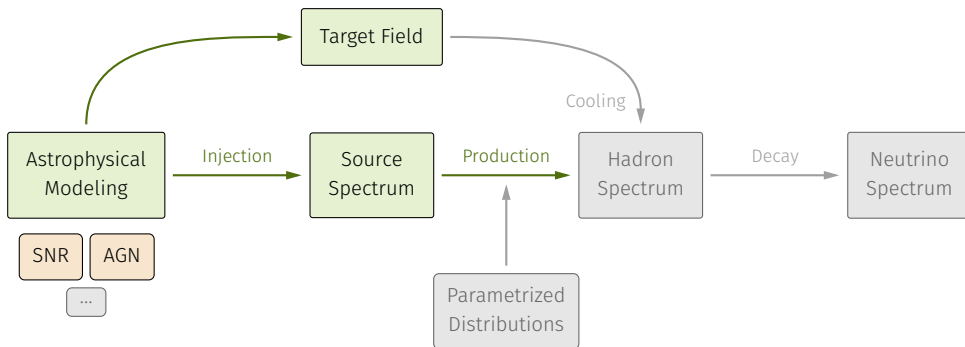
### ■ Gaussian Density Profile

$$\rho \propto \exp\left(-\frac{GM_{bh} z^2}{2u_{sn}^2 R_{dc}^3}\right)$$

### ■ Parameter Choice

$$n_{dc} = 10^{14} \text{ cm}^{-3} \qquad h = 10^{15} \text{ cm}$$







## Direct Hadron Spectrum

- Single Interaction Distribution

## Direct Hadron Spectrum

### ■ Single Interaction Distribution

$$F_{pp \rightarrow h} = E_p^{-1} \sigma_{pp}^{-1} \frac{d\sigma}{dx_E}$$

## Direct Hadron Spectrum

- Single Interaction Distribution

$$F_{pp \rightarrow h} = E_p^{-1} \sigma_{pp}^{-1} \frac{d\sigma}{dx_E}$$

- Proton Number Convolution

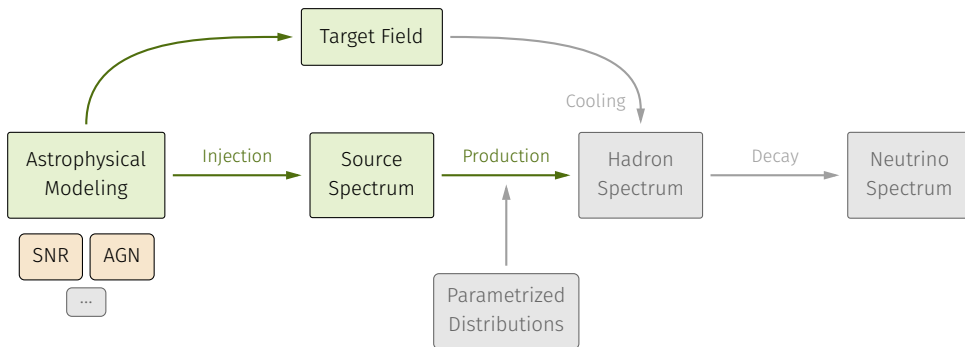
## Direct Hadron Spectrum

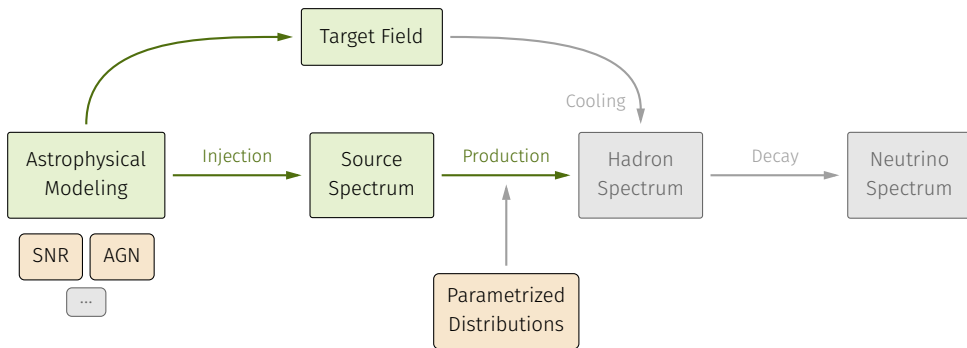
### ■ Single Interaction Distribution

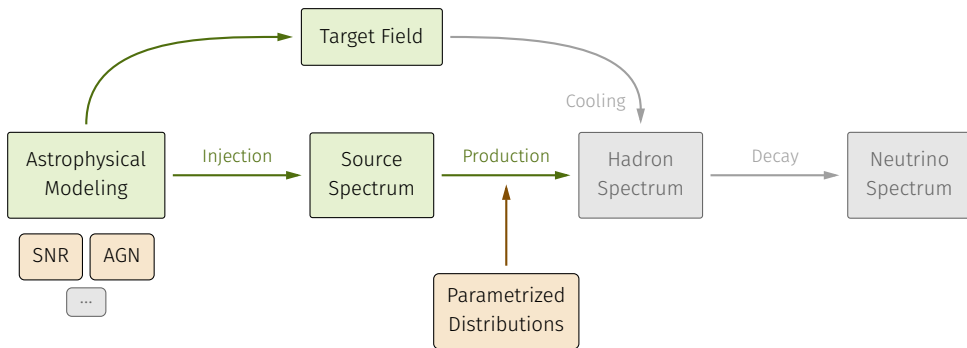
$$F_{pp \rightarrow h} = E_p^{-1} \sigma_{pp}^{-1} \frac{d\sigma}{dx_E}$$

### ■ Proton Number Convolution

$$\frac{dN_h}{dE_h} = \int_{E_h}^{\infty} dE_p \frac{dN_p}{dE_p} F_{pp \rightarrow h}$$









## Unweighted Energy Differential Cross Section

- Fragmentation Function

## Unweighted Energy Differential Cross Section

### ■ Fragmentation Function

$$D_c^h = \frac{N_h z (1 - z)^2}{((1 - z)^2 + \epsilon_h z)^2}$$

## Unweighted Energy Differential Cross Section

- Fragmentation Function

$$D_c^h = \frac{N_h z(1-z)^2}{((1-z)^2 + \epsilon_h z)^2}$$

- Charmed Hadron Spectra

## Unweighted Energy Differential Cross Section

### ■ Fragmentation Function

$$D_c^h = \frac{N_h z(1-z)^2}{((1-z)^2 + \epsilon_h z)^2}$$

### ■ Charmed Hadron Spectra

$$\frac{d\sigma}{dx_E} = \left(1 - \frac{m_h}{x_E E_p}\right)^{1/2} \int_{x_E}^1 dz z^{-1} \frac{d\sigma}{dx_c} D_c^h$$

## Unweighted Energy Differential Cross Section

### ■ Fragmentation Function

$$D_c^h = \frac{N_h z(1-z)^2}{((1-z)^2 + \epsilon_h z)^2}$$

### ■ Charmed Hadron Spectra

$$\frac{d\sigma}{dx_E} = \left(1 - \frac{m_h}{x_E E_p}\right)^{1/2} \int_{x_E}^1 dz z^{-1} \frac{d\sigma}{dx_c} D_c^h$$

## Unweighted Energy Differential Cross Section

### ■ Fragmentation Function

$$D_c^h = \frac{N_h z(1-z)^2}{((1-z)^2 + \epsilon_h z)^2}$$

### ■ Charmed Hadron Spectra

$$\frac{d\sigma}{dx_E} = \left(1 - \frac{m_h}{x_E E_p}\right)^{1/2} \int_{x_E}^1 dz z^{-1} \frac{d\sigma}{dx_c} D_c^h$$

## Unweighted Energy Differential Cross Section

### ■ Fragmentation Function

$$D_c^h = \frac{N_h z(1-z)^2}{((1-z)^2 + \epsilon_h z)^2}$$

### ■ Charmed Hadron Spectra

$$\frac{d\sigma}{dx_E} = \left(1 - \frac{m_h}{x_E E_p}\right)^{1/2} \int_{x_E}^1 dz z^{-1} \frac{d\sigma}{dx_c} D_c^h$$

### ■ Energy Fractions

## Unweighted Energy Differential Cross Section

### ■ Fragmentation Function

$$D_c^h = \frac{N_h z(1-z)^2}{((1-z)^2 + \epsilon_h z)^2}$$

### ■ Charmed Hadron Spectra

$$\frac{d\sigma}{dx_E} = \left(1 - \frac{m_h}{x_E E_p}\right)^{1/2} \int_{x_E}^1 dz z^{-1} \frac{d\sigma}{dx_c} D_c^h$$

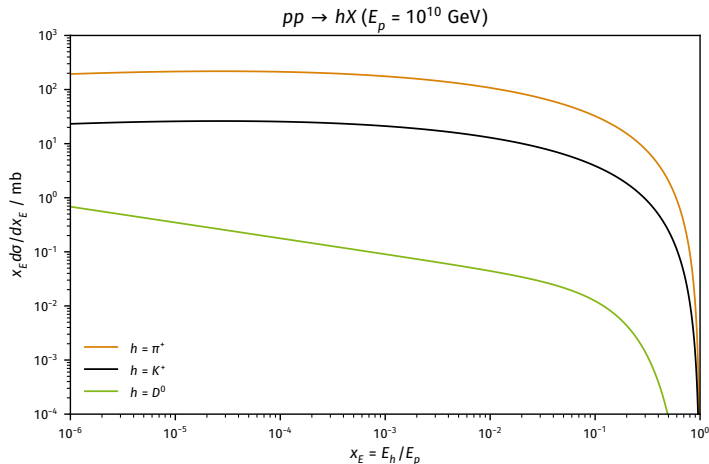
### ■ Energy Fractions

$$x_E = E_h/E_p$$

$$x_c = E_c/E_p$$

$$z = E_h/E_c$$

## Weighted Energy Differential Cross Section

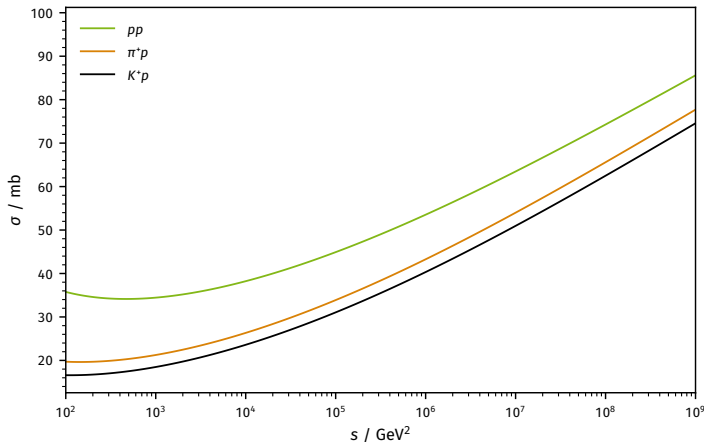


Derived Parametrizations

S. R. Kelner et al.  
*Phys. Rev. D* (2006)

V. P. Gonçalves et al.  
*JHEP* (2007)

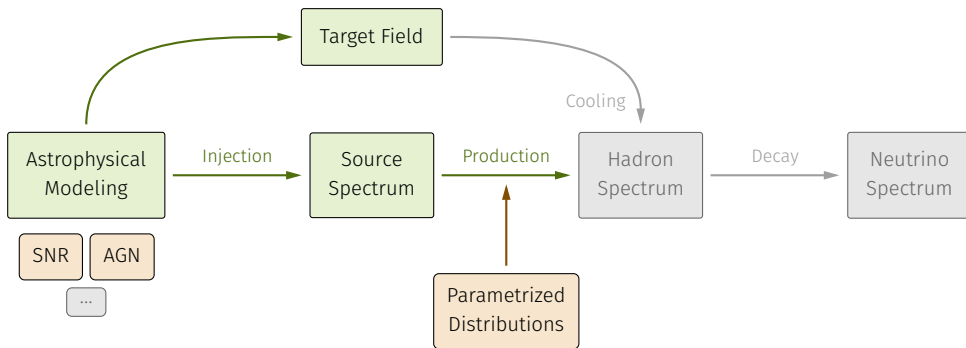
## Hadron Inelastic Scattering Cross Section

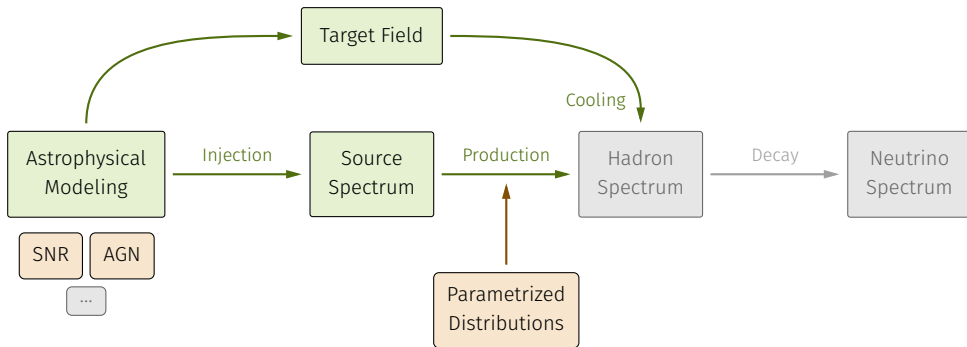


Combined Parametrizations

V. I. Belousov et al.  
*Phys. Atom. Nuclei* (2016)

D. A. Fagundes, M. J. Menon  
*Nucl. Phys. A* (2012)







## Hadronic Cooling & Decay

### ■ Characteristic Timescales

## Hadronic Cooling & Decay

- Characteristic Timescales

$$t_{cool} = (\kappa_{hp} \sigma_{hp} n c)^{-1}$$

## Hadronic Cooling & Decay

### ■ Characteristic Timescales

$$t_{cool} = (\kappa_{hp} \sigma_{hp} n c)^{-1}$$

$$t_{dec} = \tau_h E_h / m_h$$

## Hadronic Cooling & Decay

### ■ Characteristic Timescales

$$t_{cool} = (\kappa_{hp} \sigma_{hp} n c)^{-1}$$

$$t_{dec} = \tau_h E_h / m_h$$

### ■ Cooling Factor

## Hadronic Cooling & Decay

### ■ Characteristic Timescales

$$t_{cool} = (\kappa_{hp} \sigma_{hp} n c)^{-1}$$

$$t_{dec} = \tau_h E_h / m_h$$

### ■ Cooling Factor

$$C = 1 - \exp\left(-\frac{t_{cool}}{t_{dec}}\right)$$

## Hadronic Cooling & Decay

### ■ Characteristic Timescales

$$t_{cool} = (\kappa_{hp} \sigma_{hp} n c)^{-1}$$

$$t_{dec} = \tau_h E_h / m_h$$

### ■ Cooling Factor

$$C = 1 - \exp\left(-\frac{t_{cool}}{t_{dec}}\right)$$

### ■ Optical Depth

## Hadronic Cooling & Decay

### ■ Characteristic Timescales

$$t_{cool} = (\kappa_{hp} \sigma_{hp} n c)^{-1}$$

$$t_{dec} = \tau_h E_h / m_h$$

### ■ Cooling Factor

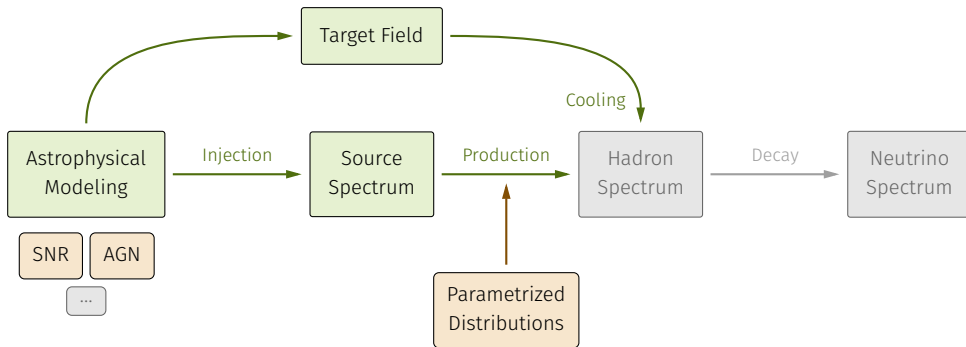
$$C = 1 - \exp\left(-\frac{t_{cool}}{t_{dec}}\right)$$

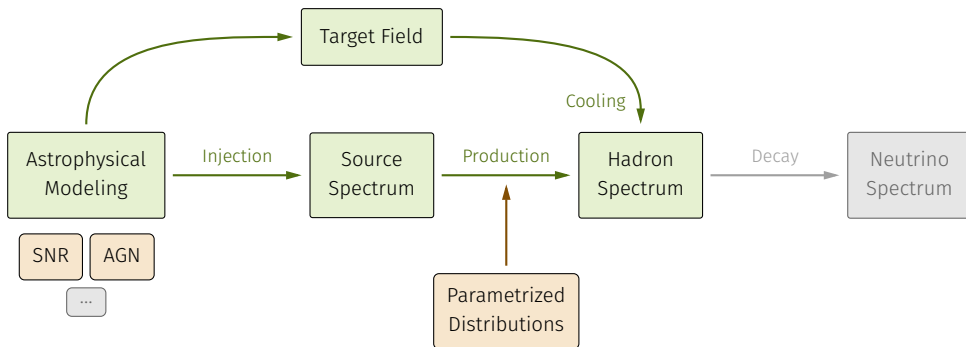
### ■ Optical Depth

$$O = \kappa_{pp} \sigma_{pp} n d$$

## Relevant Hadron Properties

| $h$           | $q_h$      | $\tau_h / \text{s}$   | $m_h / \text{GeV}$ | $s_h / \text{GeV}^2$ | $B_{h \rightarrow \nu}$ |
|---------------|------------|-----------------------|--------------------|----------------------|-------------------------|
| $p$           | $uud$      | $\infty$              | 0.94               | 15.98                | 0.000                   |
| $\pi^+$       | $u\bar{d}$ | $26.0 \times 10^{-9}$ | 0.14               | 10.23                | 1.000                   |
| $K^+$         | $u\bar{s}$ | $12.4 \times 10^{-9}$ | 0.49               | 12.62                | 0.636                   |
| $D_0$         | $c\bar{u}$ | $0.4 \times 10^{-12}$ | 1.86               | 0.45                 | 0.067                   |
| $D^+$         | $c\bar{d}$ | $1.0 \times 10^{-12}$ | 1.87               | 0.40                 | 0.176                   |
| $D_s^+$       | $c\bar{s}$ | $0.5 \times 10^{-12}$ | 1.97               | 0.71                 | 0.065                   |
| $\Lambda_c^+$ | $udc$      | $0.2 \times 10^{-12}$ | 2.29               | 1.61                 | 0.045                   |





## Produced Hadron Spectrum

### ■ Modification Factors

## Produced Hadron Spectrum

### ■ Modification Factors

$$\frac{dN_h}{dE_h} = CO \int_{E_h}^{\infty} dE_p \frac{dN_p}{dE_p} F_{pp \rightarrow h}$$

## Produced Hadron Spectrum

### ■ Modification Factors

$$\frac{dN_h}{dE_h} = CO \int_{E_h}^{\infty} dE_p \frac{dN_p}{dE_p} F_{pp \rightarrow h}$$

## Produced Hadron Spectrum

### ■ Modification Factors

$$\frac{dN_h}{dE_h} = CO \int_{E_h}^{\infty} dE_p \frac{dN_p}{dE_p} F_{pp \rightarrow h}$$

## Produced Hadron Spectrum

- Modification Factors

$$\frac{dN_h}{dE_h} = CO \int_{E_h}^{\infty} dE_p \frac{dN_p}{dE_p} F_{pp \rightarrow h}$$

- Temporal Evolution

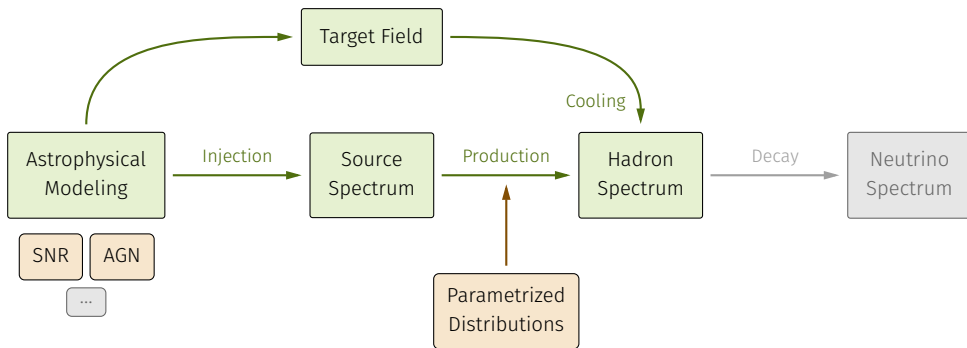
## Produced Hadron Spectrum

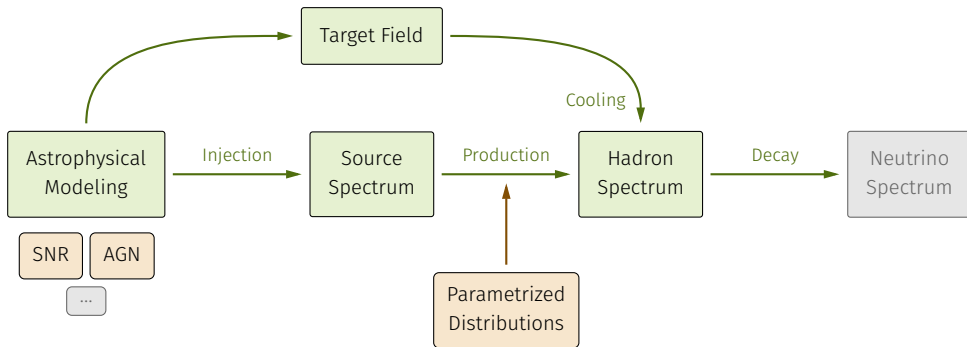
### ■ Modification Factors

$$\frac{dN_h}{dE_h} = CO \int_{E_h}^{\infty} dE_p \frac{dN_p}{dE_p} F_{pp \rightarrow h}$$

### ■ Temporal Evolution

$$CO \propto t^{-3} (1 - e^{-t^3})$$





## Resulting Neutrino Spectrum

- Complex Three Body Decay



## Resulting Neutrino Spectrum

- Complex Three Body Decay
- Simple Two Body Decay

## Resulting Neutrino Spectrum

- Complex Three Body Decay
- Simple Two Body Decay

$$\lambda_h = m_\mu^2 / m_h^2$$

## Resulting Neutrino Spectrum

- Complex Three Body Decay
- Simple Two Body Decay

$$\lambda_h = m_\mu^2 / m_h^2$$

$$F_{h \rightarrow \nu} = B_{h \rightarrow \nu} E_h^{-1} (1 - \lambda_h)^{-1}$$

## Resulting Neutrino Spectrum

- Complex Three Body Decay
- Simple Two Body Decay

$$\lambda_h = m_\mu^2 / m_h^2$$

$$F_{h \rightarrow \nu} = B_{h \rightarrow \nu} E_h^{-1} (1 - \lambda_h)^{-1}$$

- Hadron Number Convolution

## Resulting Neutrino Spectrum

- Complex Three Body Decay
- Simple Two Body Decay

$$\lambda_h = m_\mu^2 / m_h^2$$

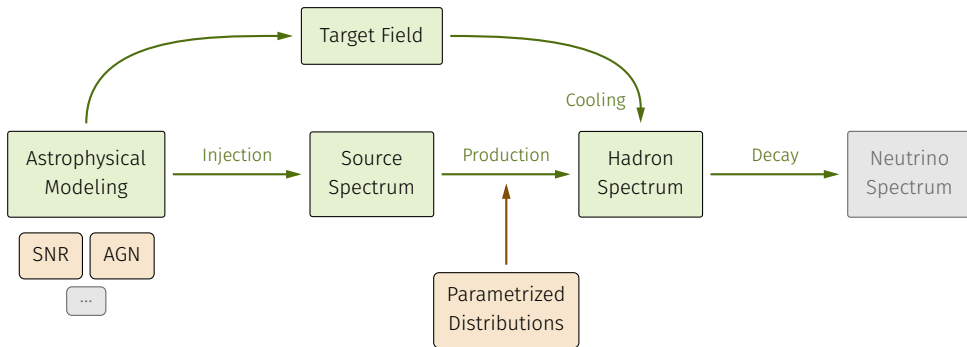
$$F_{h \rightarrow \nu} = B_{h \rightarrow \nu} E_h^{-1} (1 - \lambda_h)^{-1}$$

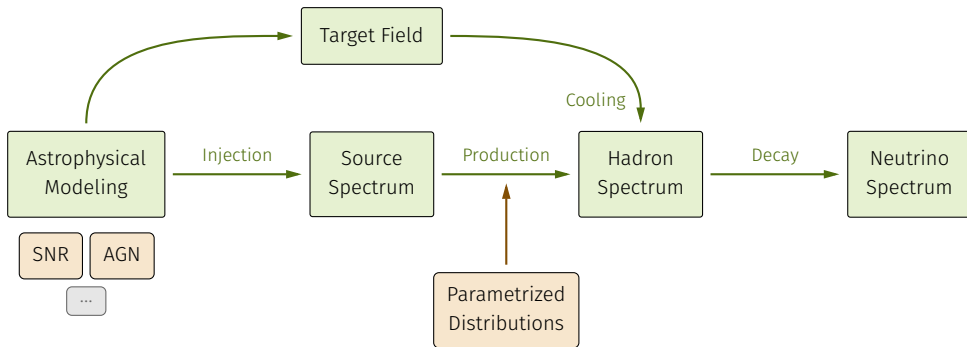
- Hadron Number Convolution

$$\frac{dN_\nu}{dE_\nu} = \int_{E_\nu}^{\infty} dE_h \frac{dN_h}{dE_h} F_{h \rightarrow \nu}$$

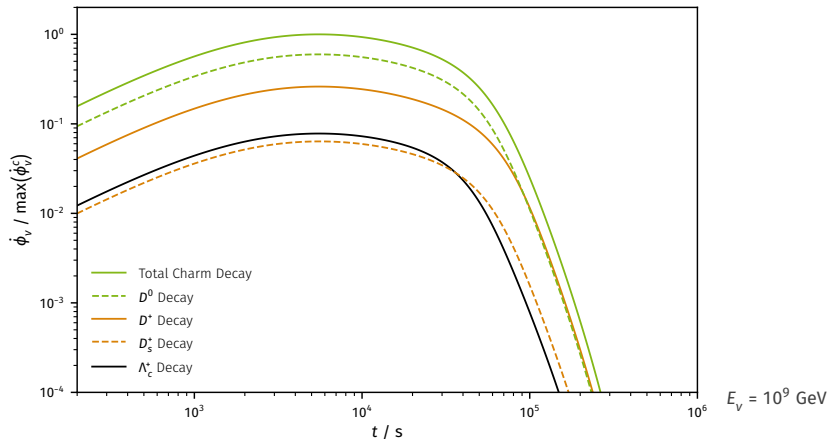
## Relevant Hadron Properties

| $h$           | $q_h$      | $\tau_h / \text{s}$   | $m_h / \text{GeV}$ | $s_h / \text{GeV}^2$ | $B_{h \rightarrow \nu}$ |
|---------------|------------|-----------------------|--------------------|----------------------|-------------------------|
| $p$           | $uud$      | $\infty$              | 0.94               | 15.98                | 0.000                   |
| $\pi^+$       | $u\bar{d}$ | $26.0 \times 10^{-9}$ | 0.14               | 10.23                | 1.000                   |
| $K^+$         | $u\bar{s}$ | $12.4 \times 10^{-9}$ | 0.49               | 12.62                | 0.636                   |
| $D_0$         | $c\bar{u}$ | $0.4 \times 10^{-12}$ | 1.86               | 0.45                 | 0.067                   |
| $D^+$         | $c\bar{d}$ | $1.0 \times 10^{-12}$ | 1.87               | 0.40                 | 0.176                   |
| $D_s^+$       | $c\bar{s}$ | $0.5 \times 10^{-12}$ | 1.97               | 0.71                 | 0.065                   |
| $\Lambda_c^+$ | $udc$      | $0.2 \times 10^{-12}$ | 2.29               | 1.61                 | 0.045                   |

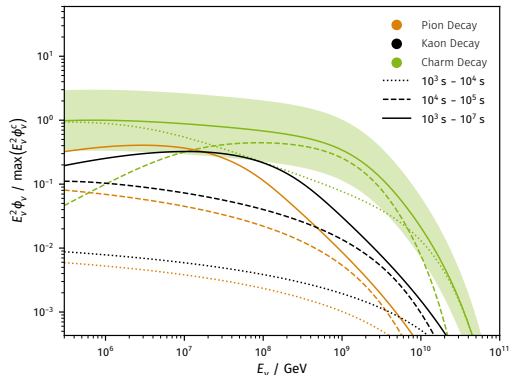
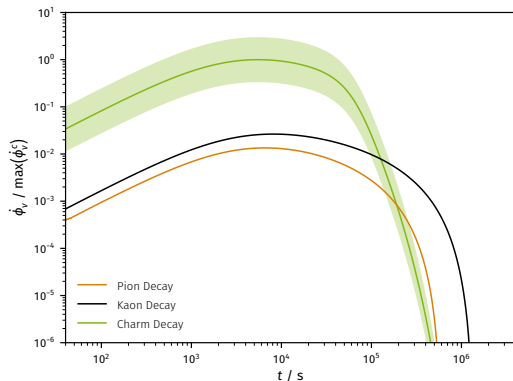




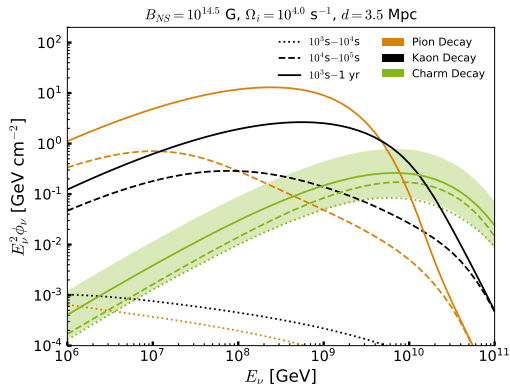
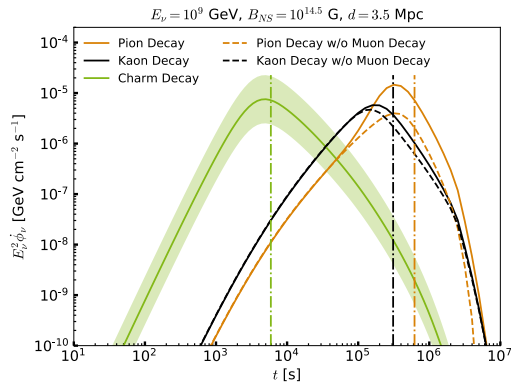
## Normalized Magnetar Charm Contributions



## Normalized Magnetar Neutrino Flux & Fluence

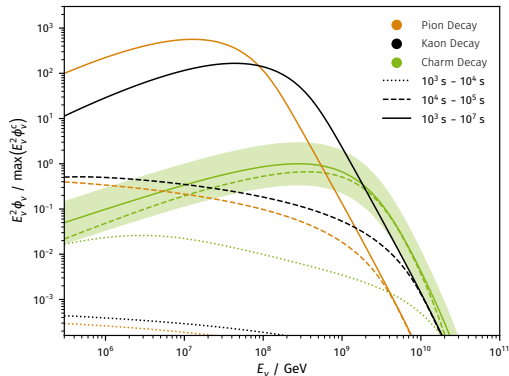
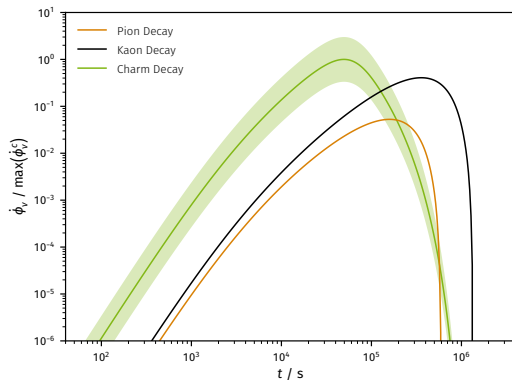


## Published Estimated Magnetar Neutrinos

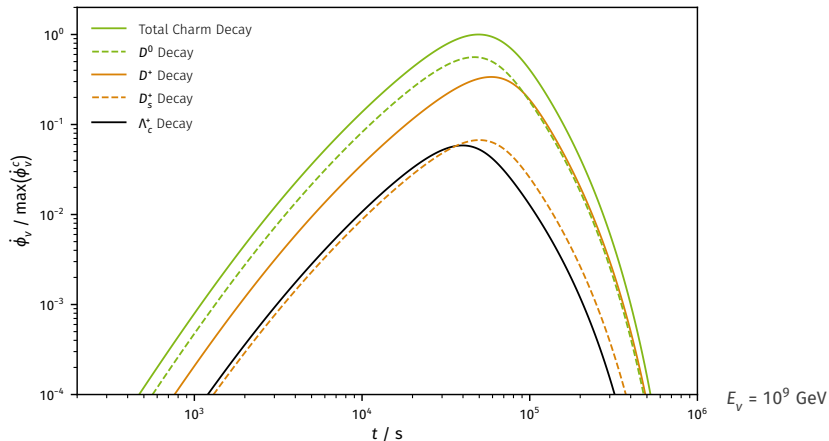


J. A. Carpio et al. *Phys. Rev. D* (2020)

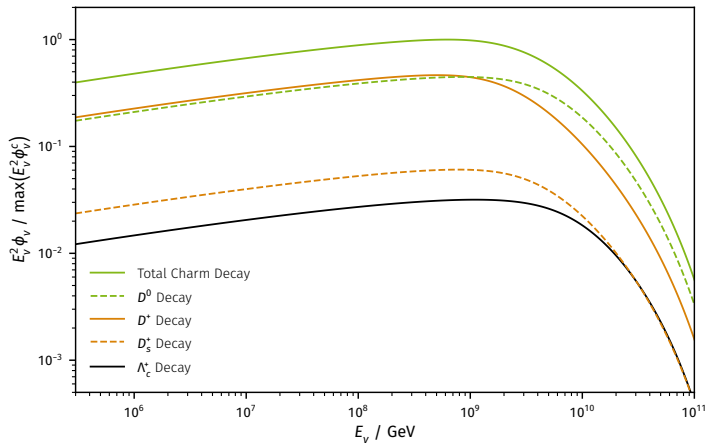
## Modified Magnetar Neutrino Flux & Fluence



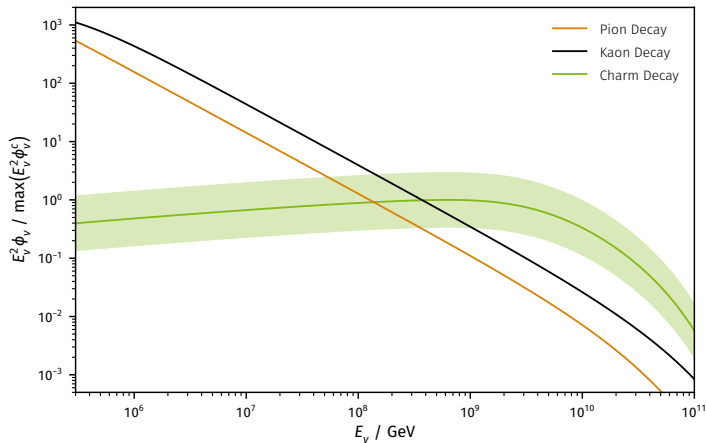
## Modified Magnetar Charm Contributions

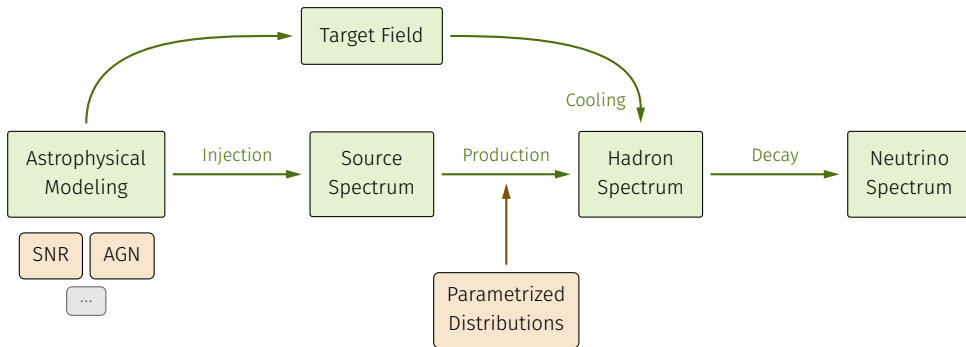


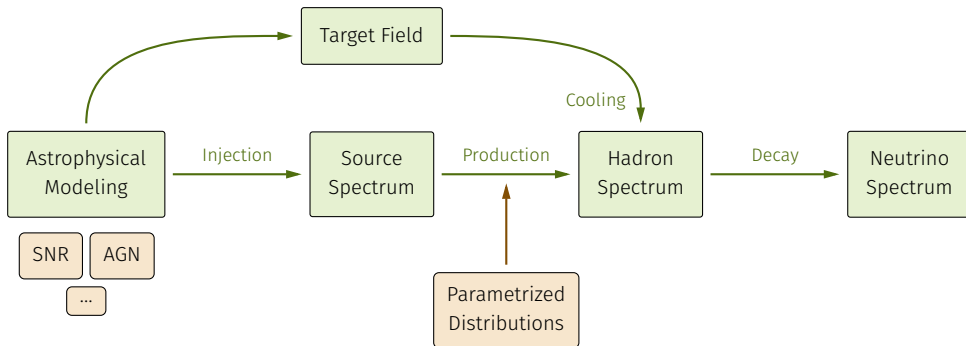
## Normalized AGN Charm Contributions



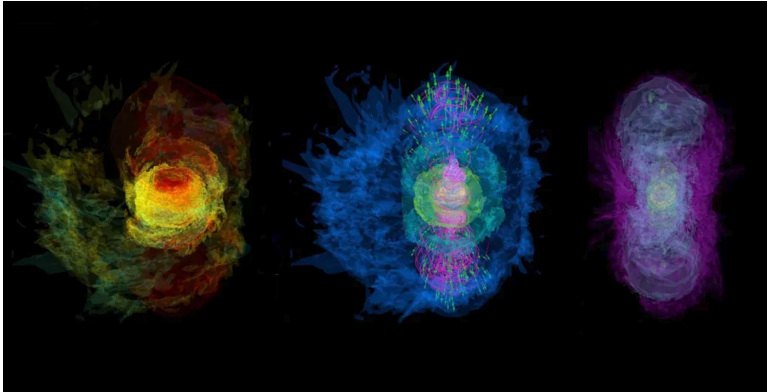
## Normalized AGN Neutrino Fluence





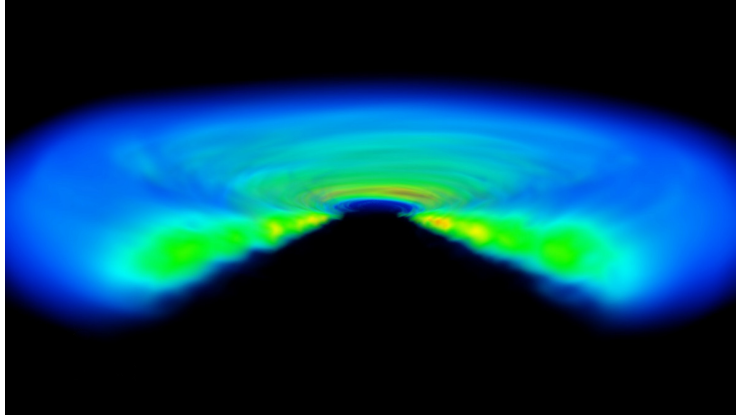


## Simulated Magnetar Formation

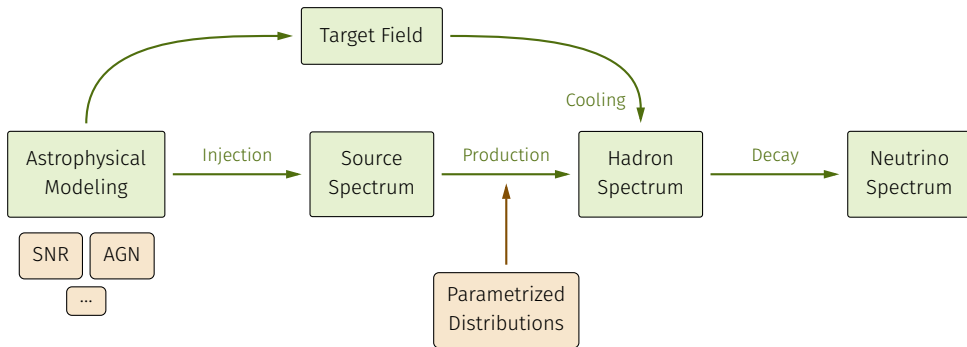


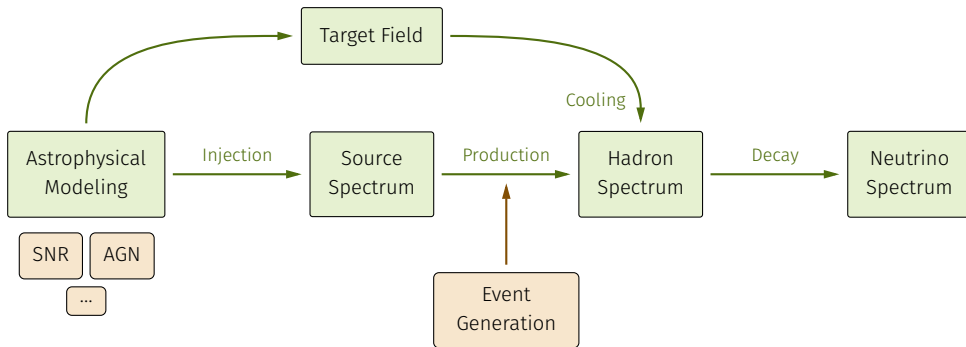
K. Hayashi *MPI Grav. Phys.* (2024)

## Accretion Disk Structure

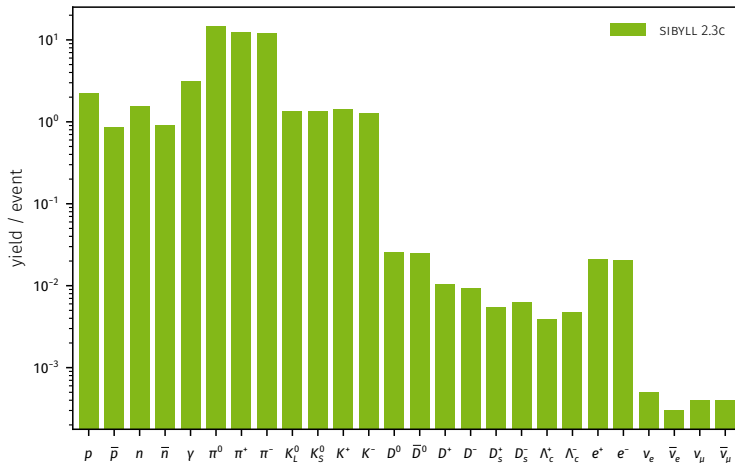


V. V. Zhuravlev et al. *ApJ* (2014)



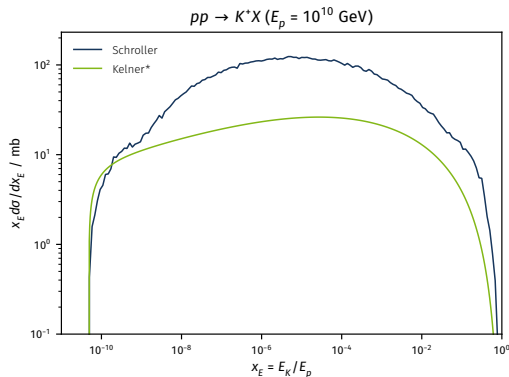
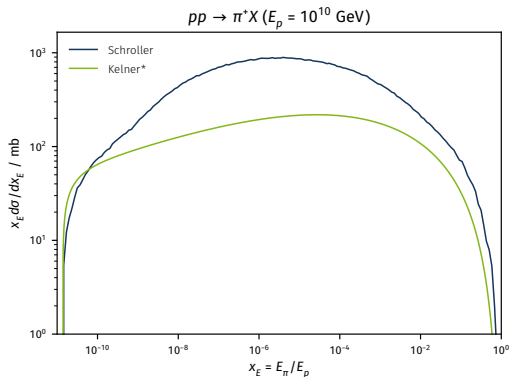


## Monte Carlo Event Generation



Data Courtesy  
M. Schroller

## Differential Cross Sections Comparison

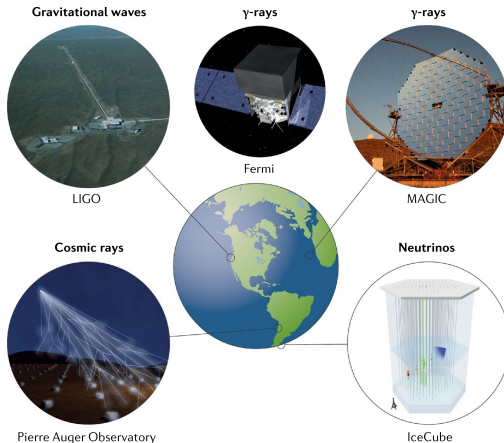


Derived Parametrization S. R. Kelner et al. *Phys. Rev. D* (2006)  
Event Data (SIBYLL 2.3c) Courtesy M. Schrollier



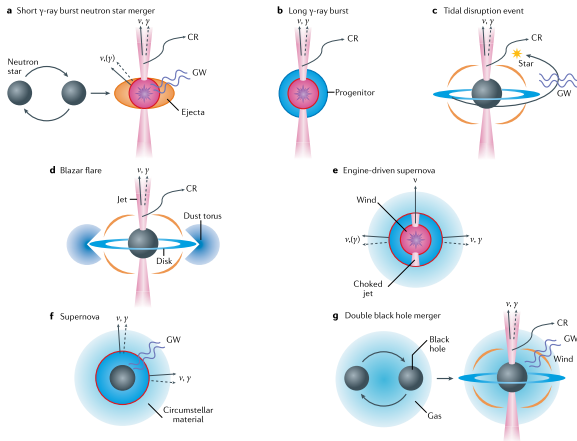
## Summary & Conclusion

## Messenger Types & Detection



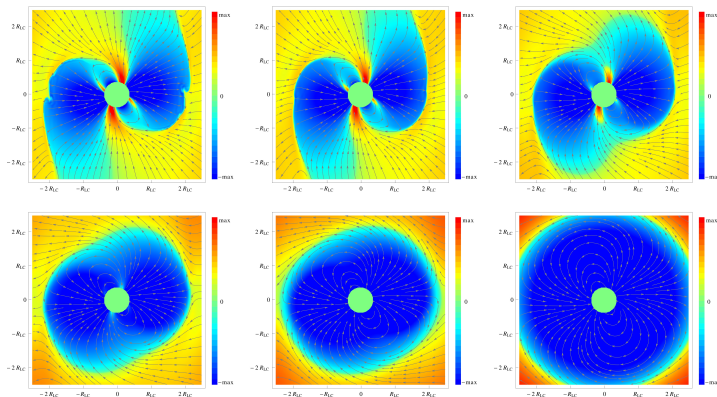
P. Mészáros et al.  
*Nat. Rev. Phys.* (2019)

## Source Regions & Events



P. Mészáros et al.  
*Nat. Rev. Phys.* (2019)

## Resistive Pulsar Magnetospheres



J. Li et al. *ApJ* (2012)

## Luminosity Models

- Vacuum Dipole Radiation

$$L = \frac{2\mu^2\omega^4\sin^2\chi}{3c^2}$$

- Force Free Magnetosphere

$$L = \frac{\mu^2\omega^4(1 + \sin^2\chi)}{c^3}$$

## Spindown Derivation

### ■ Magnetar Luminosity

$$L = K\omega^4 = -\dot{E}$$

### ■ Exponential Decay

$$E = E_0 \exp(-t/t_{sd}) \quad E = I\omega^2/2$$

$$\dot{E} = -E/t_{sd} \quad \dot{E} = I\omega\dot{\omega}$$

$$L_0 = K\omega_0^4 = I\omega_0^2/2t_{sd}$$

## Spindown Derivation

### ■ Spindown Time

$$t_{sd} = I/2K\omega_0^2$$

### ■ Differential Equation

$$I\dot{\omega}_0 = -K\omega^3$$

### ■ Separating Variables

$$\omega = \omega_0(1 + t/t_{sd})^{-1/2}$$

## Acceleration Mechanisms

### Wakefield Acceleration

$$E^M = \eta e \phi_{rc} = \eta e \psi_{rc} / R_{lc} = \eta e \mu_{ns} / R_{lc}^2$$

### Diffusive Shock Acceleration

$$N = N_0 \zeta^k \quad E = E_0 \eta^k$$

$$\frac{\ln(N/N_0)}{\ln(E/E_0)} = \frac{\ln(\zeta)}{\ln(\eta)} \quad N = N_0 \left( \frac{E}{E_0} \right)^{\ln(\zeta)/\ln(\eta)} \quad \frac{dN}{dE} = \frac{N_0}{E_0} \left( \frac{E}{E_0} \right)^{\ln(\zeta)/\ln(\eta) - 1}$$

## Hydrostatic Disk Equilibrium

### ■ Gravitational Force

$$F = \frac{GM_{bh}m_p}{R_{dc}^2}$$

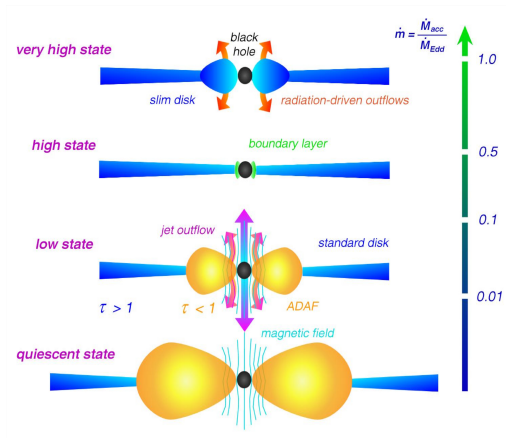
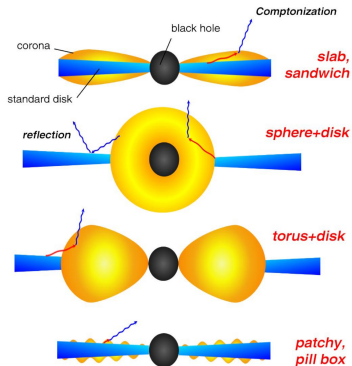
### ■ Small Angle Approximation

$$\frac{d\rho}{dz} = -\rho\ddot{z} = -\frac{\rho GM_{bh}z}{u_{sn}^2 R_{dc}^3}$$

### ■ Sound & Orbital Velocity

$$\rho = \rho_0 \exp(-z^2/2h^2) \quad h^2 = \frac{u_{sn}^2 R_{dc}^3}{GM_{bh}} = \frac{u_{sn}^2 R_{dc}^2}{v_{or}^2}$$

## Accretion Disk & Corona



A. Müller Heidelberg University (2004)

## Particle Interactions

### ■ Source Size Relation

$$R = \frac{P}{QB} \quad E < Ze\beta cBD$$

### ■ Mean Lifetime & Free Path

$$\lambda = (\kappa\sigma n)^{-1}$$

### ■ Decay & Dissipation Laws

$$P_x = 1 - \exp(-x/\lambda) \quad P_t = 1 - \exp(-t/\tau)$$

## Parametrized Distributions

### ■ Hadron Scattering

$$\sigma_{hp} = H_h \ln^2(s/s_h) + P_h + R_h^1 (s_h/s)^{\eta_1} + R_h^2 (s_h/s)^{\eta_2}$$

### ■ Charm Cross Section

$$x_F = p_c/p_s \quad x_F \frac{d\sigma}{dx_F} = a x_F^b (1 - x_F^m)^n$$

$$a = a_1 \ln(E_p) - a_2 \quad b = b_1 - b_2 \ln(E_p) \quad n = n_1 - n_2 \ln(E_p)$$

## Parametrized Distributions

### ■ Neutral Pion Distribution

$$F_{pp \rightarrow \pi} = 4E_p^{-1} \alpha B x_\pi^{\alpha-1} \left( \frac{y_\pi}{1 + r x_\pi^\alpha y_\pi} \right)^4 \left( y_\pi^{-1} + \frac{r(1 - 2x_\pi^\alpha)}{1 + r x_\pi^\alpha y_\pi} \right) \left( 1 - \frac{m_\pi}{x_\pi E_p} \right)^{1/2}$$

$$y_\pi = 1 - x_\pi^\alpha \quad B = \tilde{B} + C \quad \alpha = \tilde{\alpha} / \sqrt{C} \quad r = \tilde{r} / \sqrt{C}$$

$$C = c_1 - c_2 \ln(E_p) + c_3 \ln^2(E_p)$$

## Parametrized Distributions

### ■ Effective Three Body Decays

$$F_{h \rightarrow \nu} = B_{h \rightarrow \nu} E_h^{-1} D_h^{-1} \left( 6b_h a_h^2 - 4a_h^3 - 12\lambda_h^2 a_h + 12\lambda_h^2 y - 6b_h y^2 + 4y^3 + 12\lambda_h^2 \ln((1-y)/\lambda_h) \right)$$

$$y = E_\nu / E_h \quad \lambda_h = s_h / m_h^2 \quad a_h = 1 - \lambda_h \quad b_h = 1 - 2\lambda_h$$

$$D_h = 1 - 8\lambda_h - 12\lambda_h^2 \ln(\lambda_h) + 8\lambda_h^3 - \lambda_h^4$$